

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 2

Year: 2014

Subject: 340152 Electrical Measuring Instruments

Section: 5-6

Date: 10 March 2015

Time: 15.00-17.00

Name: _____ ID: _____ Class: _____

Instructions:

1. Cheating will result in all classes registered for the current semester. Student who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. This will be a closed book test.
4. Calculation devices are allowed.
5. Dictionaries are NOT permitted.
6. Electronic communication device are NOT allowed in the examination room.
7. The examination has 7 page (including this page), 13 questions total score of 70 points

Q1) Convert the following numbers as the required unit (3 marks)

- a) Convert 0.035 A to milliamperes
- b) Convert 10 pF to microfarads
- c) Convert 20 kHz to megahertz

Q2) Identify each of the instrument symbols illustrated in Figures. (4 marks)



a)



b)

Class 1.5



c)

d)

Q3) Define the following terms:

(4 marks)

(a) Error :

(b) Accuracy :

(c) Precision :

(d) Resolution :

Q4) Two resistors, R_1 and R_2 are connected in series with the following values $R_1 = 103.5 \, \Omega$ and $R_2 = 28.26 \, \Omega$. Calculate the total resistance with *four significant figures*. (2 marks)

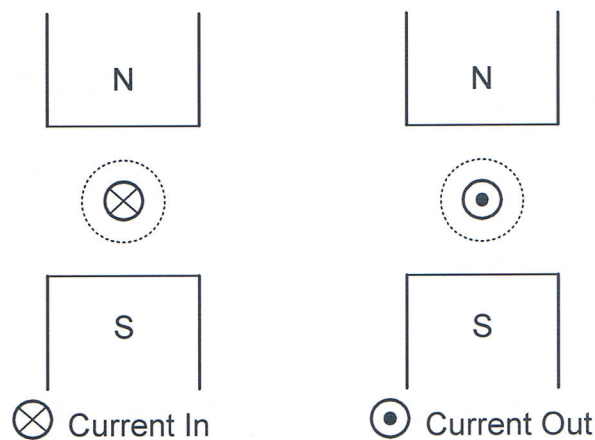
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Q5) Explain Gross, Systematic and Random Errors.

(3 marks)

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Q6) Using *Right Hand Grip Rule* to sketch the direction of the *magnetic field* and *force*. (3 marks)



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Q8) Explain the characteristic of PMMC scale (linear or nonlinear). (3 marks)

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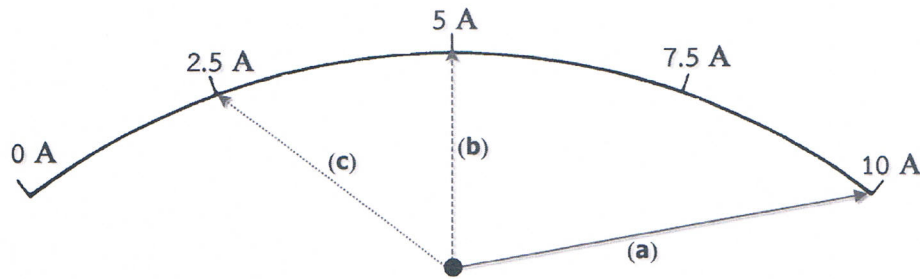
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Q9) The expected value of a resistor is $150\text{ k}\Omega$, but the measurement using ohmmeter yields a value of $149.4\text{ k}\Omega$. Calculate (a) **absolute error**, (b) **% error**, and (c) **% of accuracy**. (5 marks)

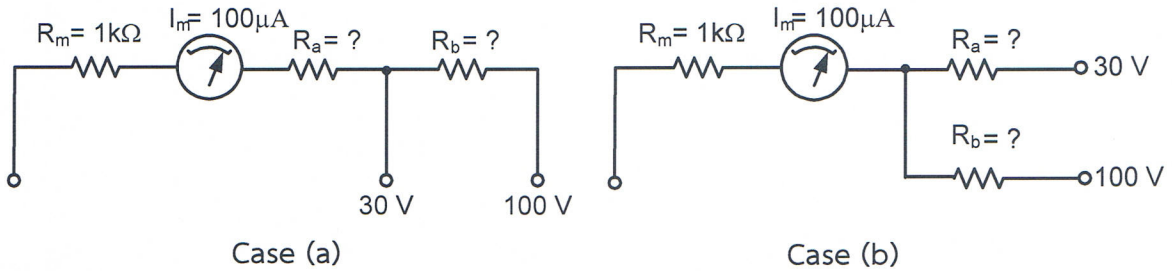
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Q10) An Ammeter range 10A class 2.5, Calculate the maximum % error for 3 different position of pointer case (a) = 10A, case (b) = 5A, case (c) = 2.5A, (10 marks)



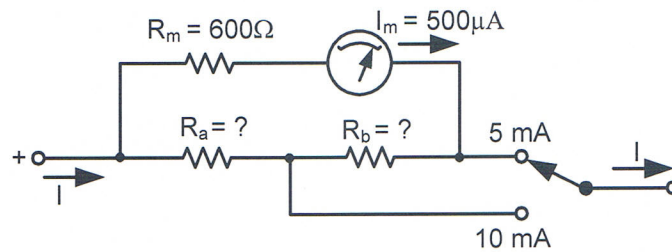
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Q11) A PMMC instrument with FSD = $100\ \mu\text{A}$ and $R_m = 1\text{k}\Omega$, is to be employed as **a volt-meter** with **ranges of 30 V and 100 V**. **Calculate** the required values of multiplier resistors (R_a and R_b) for both cases. Give the **comparison** to the practical volt-meter. (10 marks)



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Q12) A PMMC instrument has $500\ \mu\text{A}$ and a coil resistance of $600\ \Omega$. Calculate the required resistance (R_a , R_b) values to convert the instrument into *an ammeter* (multi-range) with an FSD of (a) $5\ \text{mA}$, (b) $10\ \text{mA}$ (10 marks)



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